

BIOL 220 Problem Set 05: Null Hypothesis Significance Testing

Answer Key

Identifying H_0 and H_A

💡 Questions 1–3 will get you thinking about what distinguishes null and alternative hypotheses

1. Give an appropriate null hypothesis for this alternative hypothesis [1 point]:

H_A : The mean leaf size of plants from tropical latitudes is different than the mean leaf size of plants from temperate latitudes.

Possible answer

H_0 : The mean leaf size of plants from tropical latitudes is equal to the mean leaf size of plants from temperate latitudes

The probability is the area under the curve, which is greatest between 8 and 10 in this example.

2. Give an appropriate alternative hypothesis for this null hypothesis [1 point]:

H_0 : There is no difference in the mean blood concentrations of C-reactive protein (used to diagnose Crohn's disease, units: mg/dL) in patients given a diet high in broccoli sprouts compared to the control group

Possible answer

H_A : The mean blood concentrations of C-reactive protein (used to diagnose Crohn's disease, units: mg/dL) in patients given a diet high in broccoli sprouts is different to the mean blood concentrations of C-reactive protein to the control group

3. Determine whether each hypothesis below is more appropriate as a null hypothesis or an alternative hypothesis: [1 point]

- Hypothesis: sea water temperature changes the proportion of corals that experience bleaching.
- Hypothesis: the ratio of body mass to brain mass across mammals equals 1.
- Hypothesis: the average number of animal species within tropical latitudes is the same as the average number of animal species outside of the tropics
- Hypothesis: the effect of a new experimental drug called “feel better syrup” has no effect on duration of common cold symptoms

Answer

- Hypothesis: sea water temperature changes the proportion of corals that experience bleaching. H_A
- Hypothesis: the ratio of body mass to brain mass across mammals equals 1. H_0
- Hypothesis: the average number of animal species within tropical latitudes is the same as the average number of animal species outside of the tropics H_A
- Hypothesis: the effect of a new experimental drug called “feel better syrup” has no effect on duration of common cold symptoms H_0

P-values, sample size, and whether the null is really true

💡 Pay close attention to the details in questions 4–5!

- Consider a clinical trial of a new drug to treat Alzheimer’s. Half the participants are given the drug; half are given a placebo control. Treatments are assigned at random. Assuming that the drug actually has no effect, which of the following statements is true? [1 point]
 - A clinical trial with a larger sample is **more** likely than a smaller trial to get a result with $p < 0.05$
 - A clinical trial with a larger sample is **less** likely than a smaller trial to get a result with $p < 0.05$
 - A clinical trial with a larger sample is **equally** likely than a smaller trial to get a result with $p < 0.05$
 - None of the above are true

Answer

- c. A clinical trial with a larger sample is **equally** likely than a smaller trial to get a result with $p < 0.05$

This may seem counter-intuitive but the reason is that when there truly is no effect (i.e. when H_0 really is true), then the chance of getting $p < 0.05$ is the same across sample sizes

5. Consider a clinical trial of a new drug to treat Alzheimer's. Half the participants are given the drug; half are given a placebo control. Treatments are assigned at random. Assuming that the drug *actually does reduce the chance of developing Alzheimer's*, which of the following statements is true? [1 point]
- a. A clinical trial with a larger sample is **more** likely than a smaller trial to get a result with $p < 0.05$
 - b. A clinical trial with a larger sample is **less** likely than a smaller trial to get a result with $p < 0.05$
 - c. A clinical trial with a larger sample is **equally** likely than a smaller trial to get a result with $p < 0.05$
 - d. None of the above are true

Answer

- a. A clinical trial with a larger sample is **more** likely than a smaller trial to get a result with $p < 0.05$